Finding new sites for Fuels to Flows

Jim Pivarski

THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

OCCIDENTAL ARTS & ECOLOGY CENTER

Fuels to Flows: solving two problems at once

Removing ladder fuels that cause forest fires.

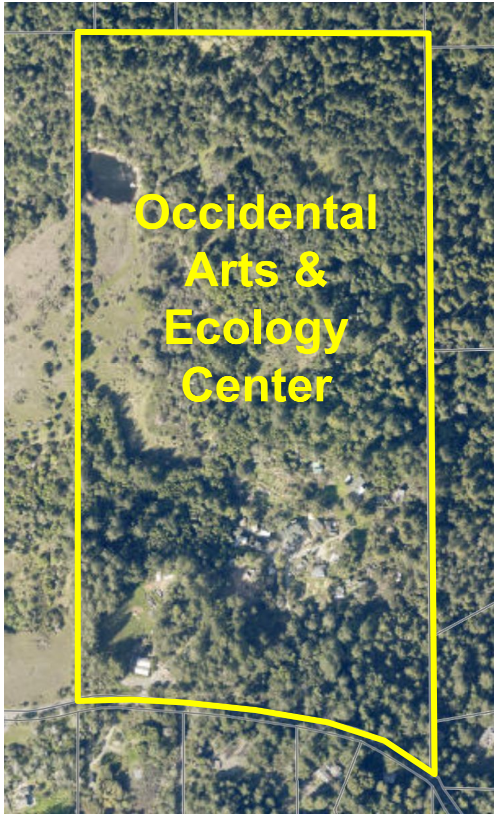
Stuffing gullies to decrease erosion.

Fuels to Flows: solving two problems at once

Before (Oct. 2021)

After (Oct. 2021)

Expanding the Fuels to Flows program

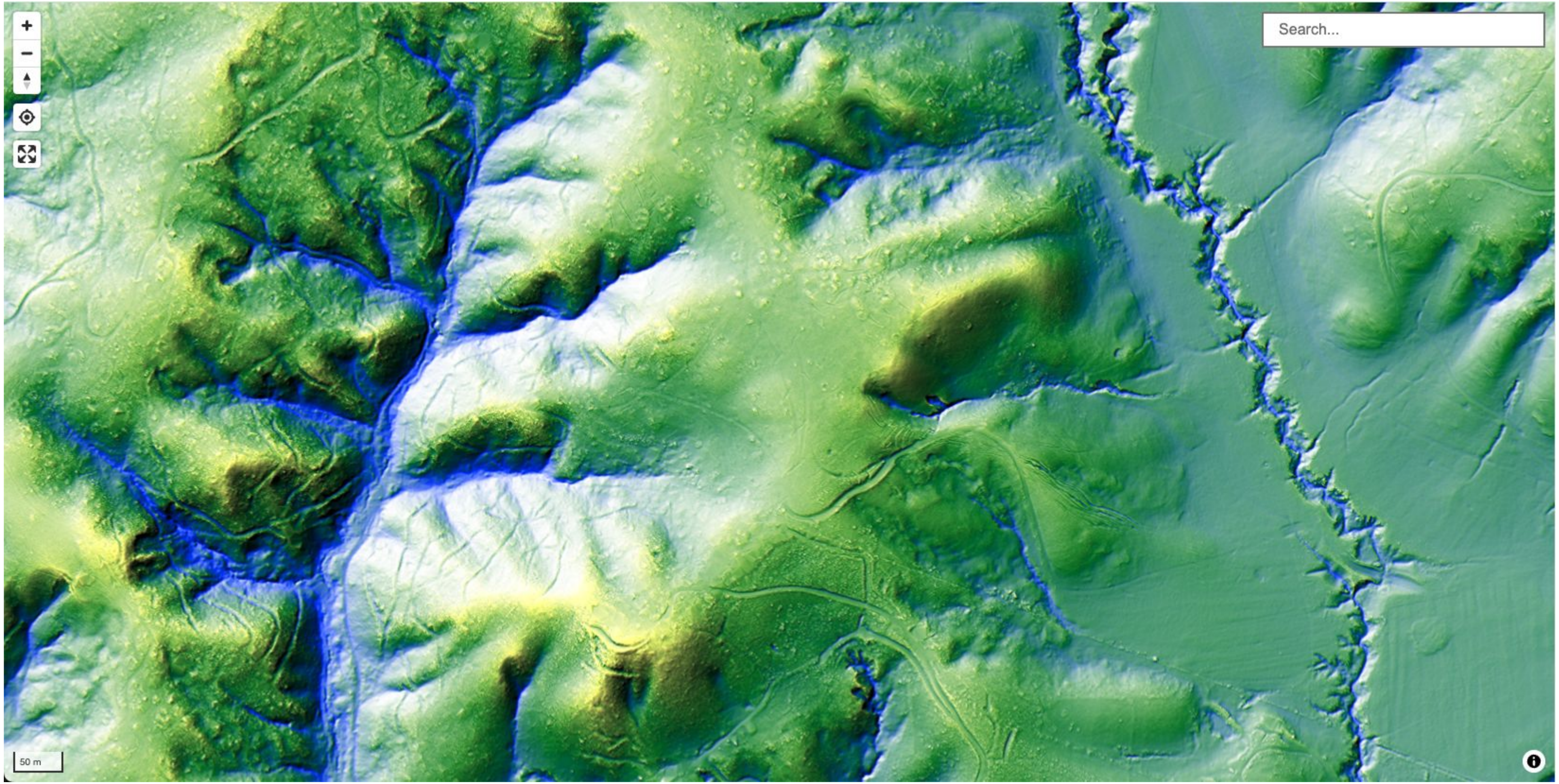


- Fuels to Flows has already been implemented at the Occidental Arts & Ecology Center (OAEC) and Monte Rio National Park.
- Program Director **Brock Dolman** was inspired by the use of LIDAR to find lost meadows and proposed a "found gully" project to find new sites for Fuels to Flows by air.
- That's how we at the DSI got involved: we developed a model to find gullies and as well as other tools to discover new Fuels to Flows sites.

Finding gullies

Where are the gullies? — aerial photography

Where are the gullies? — LIDAR hillshade without trees



Where are the gullies? — our gully-finding algorithm

Publicly available LIDAR for all of Sonoma County, CA

ABOUT | LIBRARY | DOWNLOAD DATA | VEG MAP NEWS | BLOGS | STORY MAPS, WEBMAPS & APPS

SONOMA VEG MAP
SONOMA COUNTY VEGETATION MAPPING & LIDAR PROGRAM
High-Quality Data for Planning, Conservation and Resource Management

Data Downloads

2022 LiDAR Products and Derivatives

Sonoma County Ag + Open Space and Sonoma Water, together with the North Coast Resource Partnership, Humboldt County, and a broad coalition of state and federal agencies, academic institutions, and regional stakeholders, are proud to announce the public release of the Sonoma Countywide 2022 Lidar Products. These high-resolution datasets are derived from lidar acquired between September and November 2022 as part of a larger Northern California regional collaboration and are now accessible for public use.

Visit the North Coast Resource Partnership story map for additional details
>> Story Map

Please see the datasheet for the full list of 2022 lidar derivatives available for download through Pacific Veg Map. In addition, see the table below for download/access options for a select set of data.

2022 DATA PRODUCT	2022 DATA PRODUCT LOCATION OR ENDPOINT
1-meter Bare Earth DEM	Download TIF, Service Endpoint (NAD83 State Plane), Service Endpoint (Web Mercator)
1-meter Bare Earth DEM Hillshade	Download TIF, Service Endpoint (NAD83 State Plane), Service Endpoint (Web Mercator)
1-meter Digital Surface Model/Highest Hit	Download TIF, Service Endpoint (NAD83 State Plane), Service Endpoint (Web Mercator)

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Contributing Partners

- Digital Elevation Model (DEM)
    - absolute elevation above sea level
    - in a grid of pixels: ½ meter × ½ meter
    - complete map in 2013 and 2022
- Ladder fuel proxies
    - (material in 1–4 m) / (material in 0–4 m)
    - (material in 1–8 m) / (material in 0–8 m)

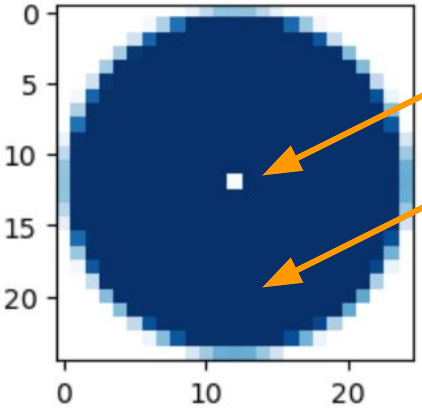
Standard technique: difference from mean elevation

Red: known gullies
on OAEC site

- For example, DOI 10.1002/esp.1918.
- A gully is an indentation in the ground.
  - Subtract the absolute elevation of each pixel from the average in a 50 meter radius.
- Reliably highlights the locations of known gullies on the OAEC site.

Note that procedure is a convolution!

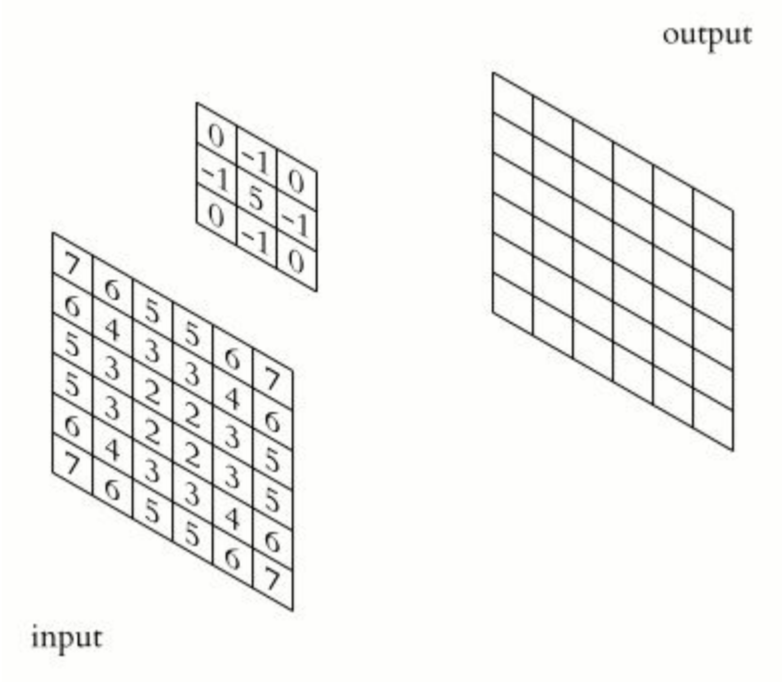
Convolution kernel:



Positive central pixel

Negative disk

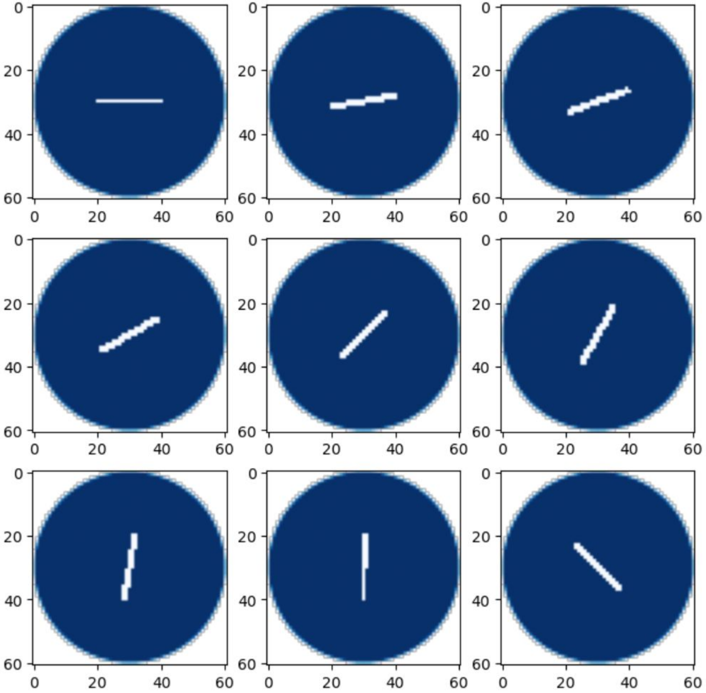
Sum = 0

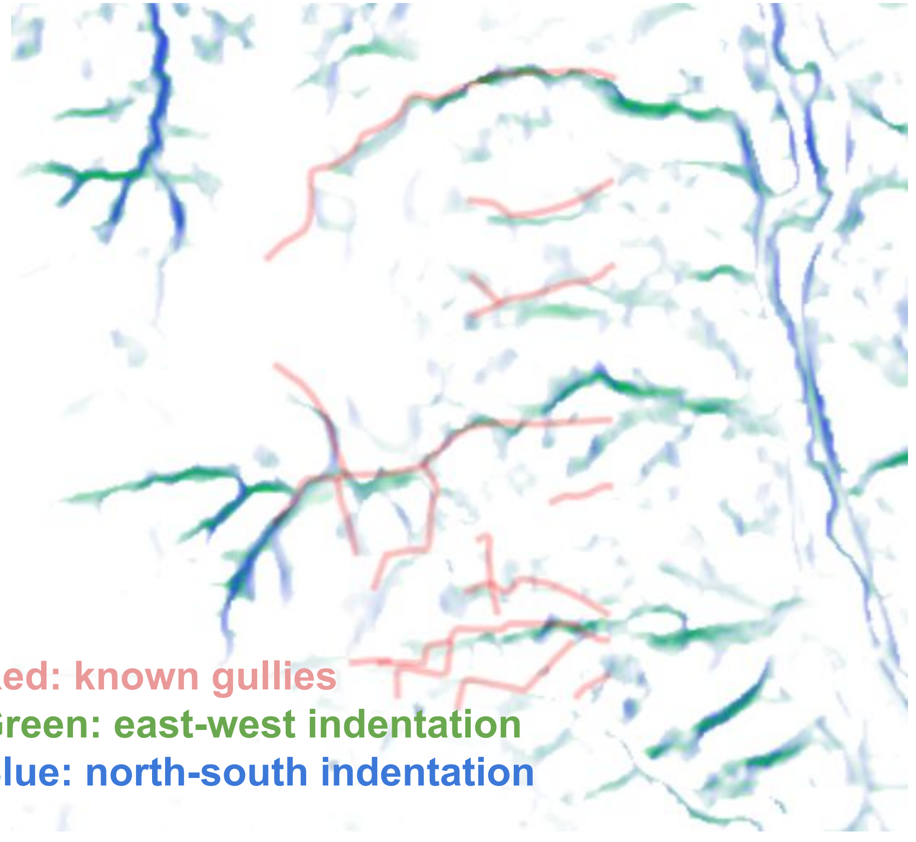


Convolution animation from Wikipedia

What if we used a convolution kernel *shaped* like a gully?

Convolution kernels:



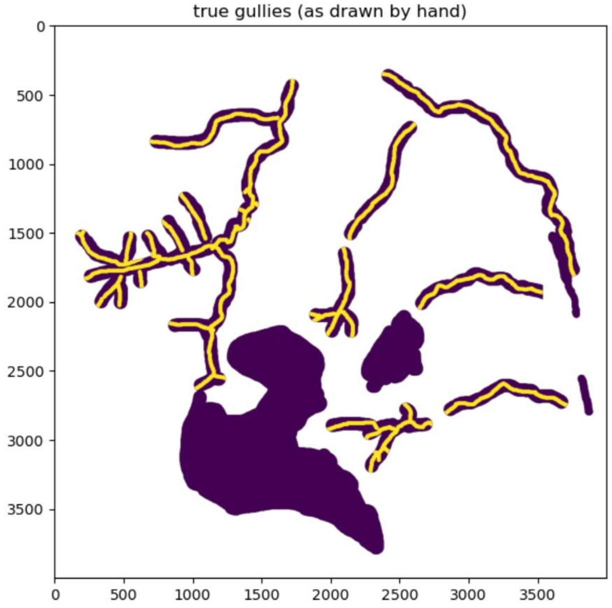


Developing a model

- This reasoning ultimately leads to a Convolutional Neural Network (CNN).
  - Rather than building a set of convolution kernels by hand, they'd be learned.
  - As a supervised learning technique, a CNN would require hand-labeled training data.

- We can label some gullies by hand, but not enough to train a CNN.

- Compromise:
  - We hand-draw gully labels for training data (in-gully, out-of-gully, and unknown) in 9 watersheds.
  - We engineer convolution kernels as features (different segment lengths; anti-cliff veto).
  - Perform logistic regression, which is equivalent to a single CNN layer (with sigmoid activation).

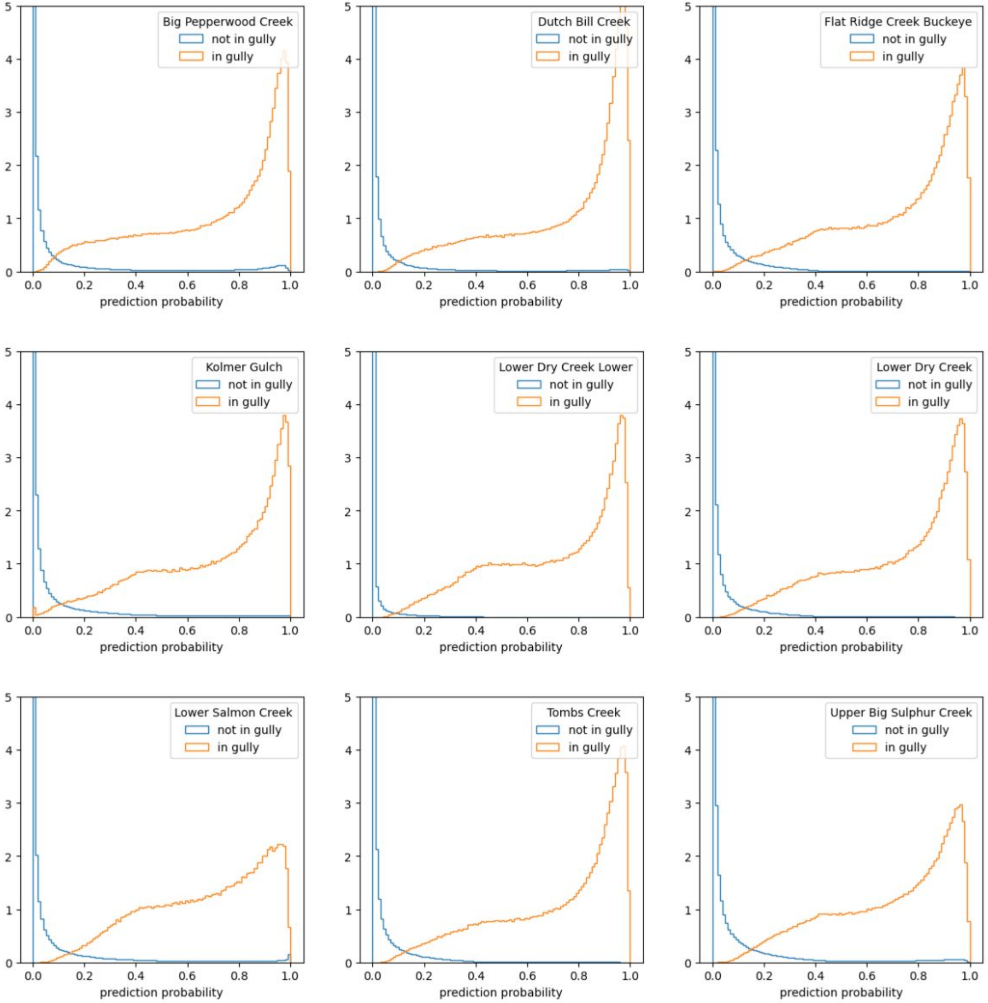


Gully probability model

- Just 7 features
- Predicts billions of pixels
- Low correlation among the features:

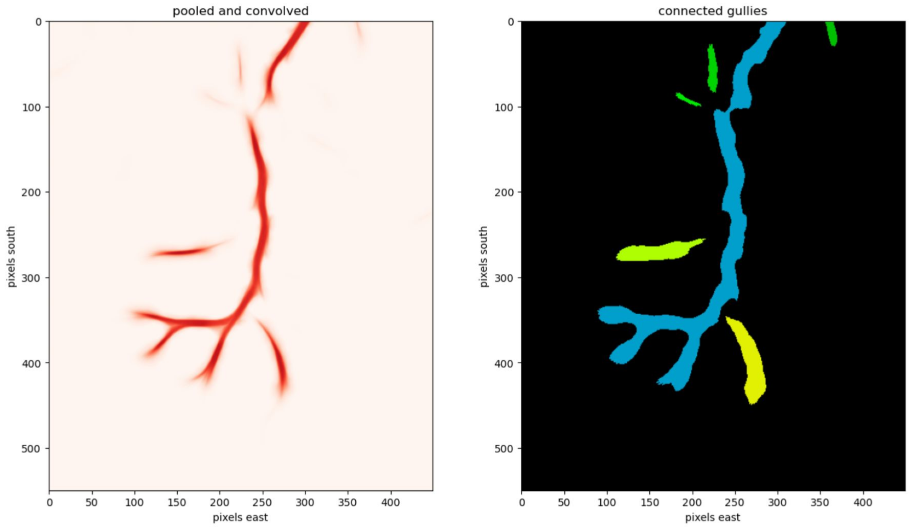
	(intercept)	min15	low15 - min15	high15 - low15	min5 - min15	high5 - low5	low15 * (high15 - low15)	abs(mindisk)
(intercept)	1.000000	0.854740	0.086704	-0.472281	-0.000096	-0.011087	-0.825057	0.029504
min15	0.854740	1.000000	0.261732	-0.094373	-0.297184	-0.241753	-0.738537	0.285971
low15 - min15	0.086704	0.261732	1.000000	0.219852	-0.262007	-0.499590	-0.057264	0.130928
high15 - low15	-0.472281	-0.094373	0.219852	1.000000	-0.625818	-0.619049	0.643569	0.038105
min5 - min15	-0.000096	-0.297184	-0.262007	-0.625818	1.000000	0.473387	-0.190269	-0.150616
high5 - low5	-0.011087	-0.241753	-0.499590	-0.619049	0.473387	1.000000	-0.076208	-0.095644
low15 * (high15 - low15)	-0.825057	-0.738537	-0.057264	0.643569	-0.190269	-0.076208	1.000000	-0.127800
abs(mindisk)	0.029504	0.285971	0.130928	0.038105	-0.150616	-0.095644	-0.127800	1.000000

- Minimized loss function →

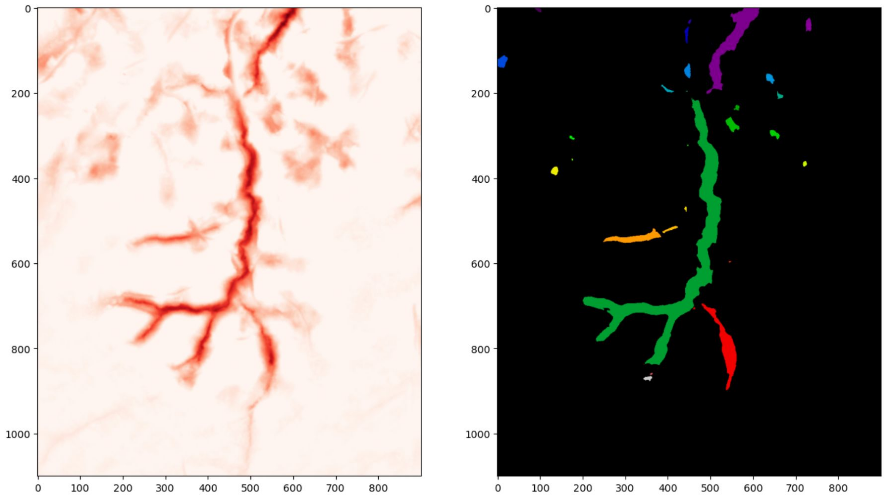


Second layer of the "CNN"

- To reduce noise and increase connectivity between real gullies:
  - Pooling: scale gully probability image down by a factor of 2 on each side.
  - Second convolution: use angled kernels again to connect islands.



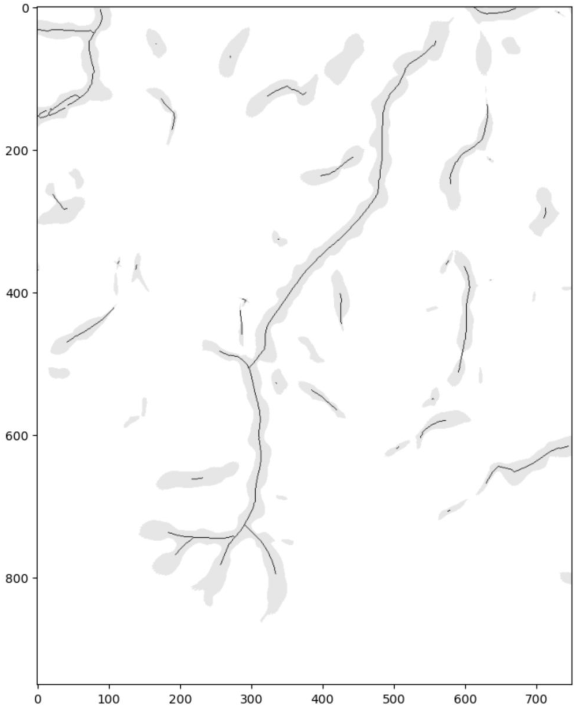
Our model



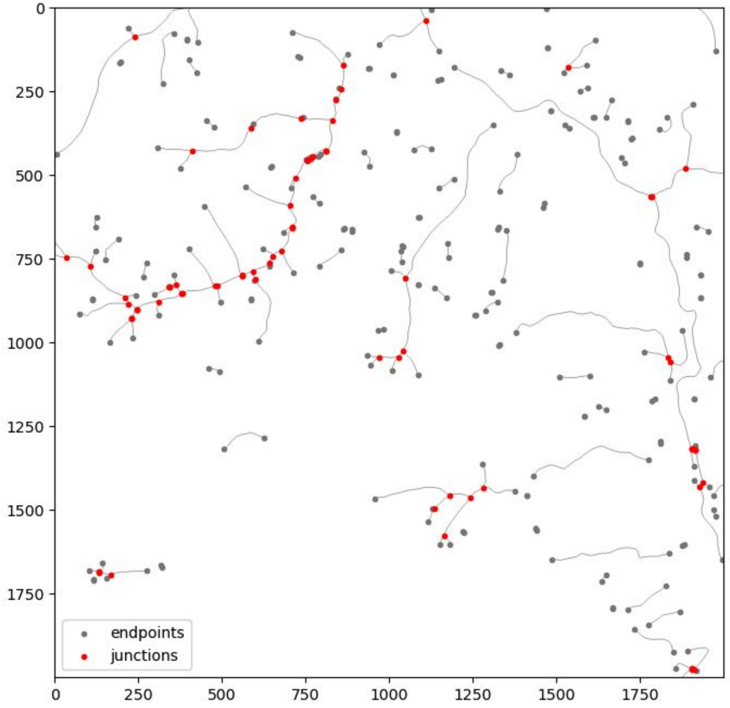
Status quo (diff from mean elevation)

Final output: a road network/graph

1. Shrink to centerlines.

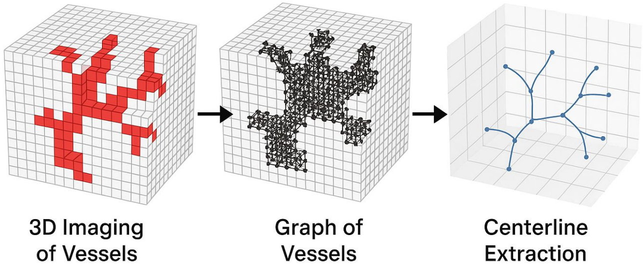


2. Connect segments into a graph.



Synergy with the DSI Clinic

- José Cardona is a masters student in Computational Analysis & Public Policy who took the DSI Clinic in Fall '25.
- His project team was analyzing the graph structure of blood vessels in breast cancer MRIs, advised by Anna Woodard and Jim Pivarski (me).
- He generalized the centerline → graph algorithm to 3D so that it could be applied to get a 3D graph of blood vessels, rather than a 2D graph of gullies.



Delivering results to OAEC

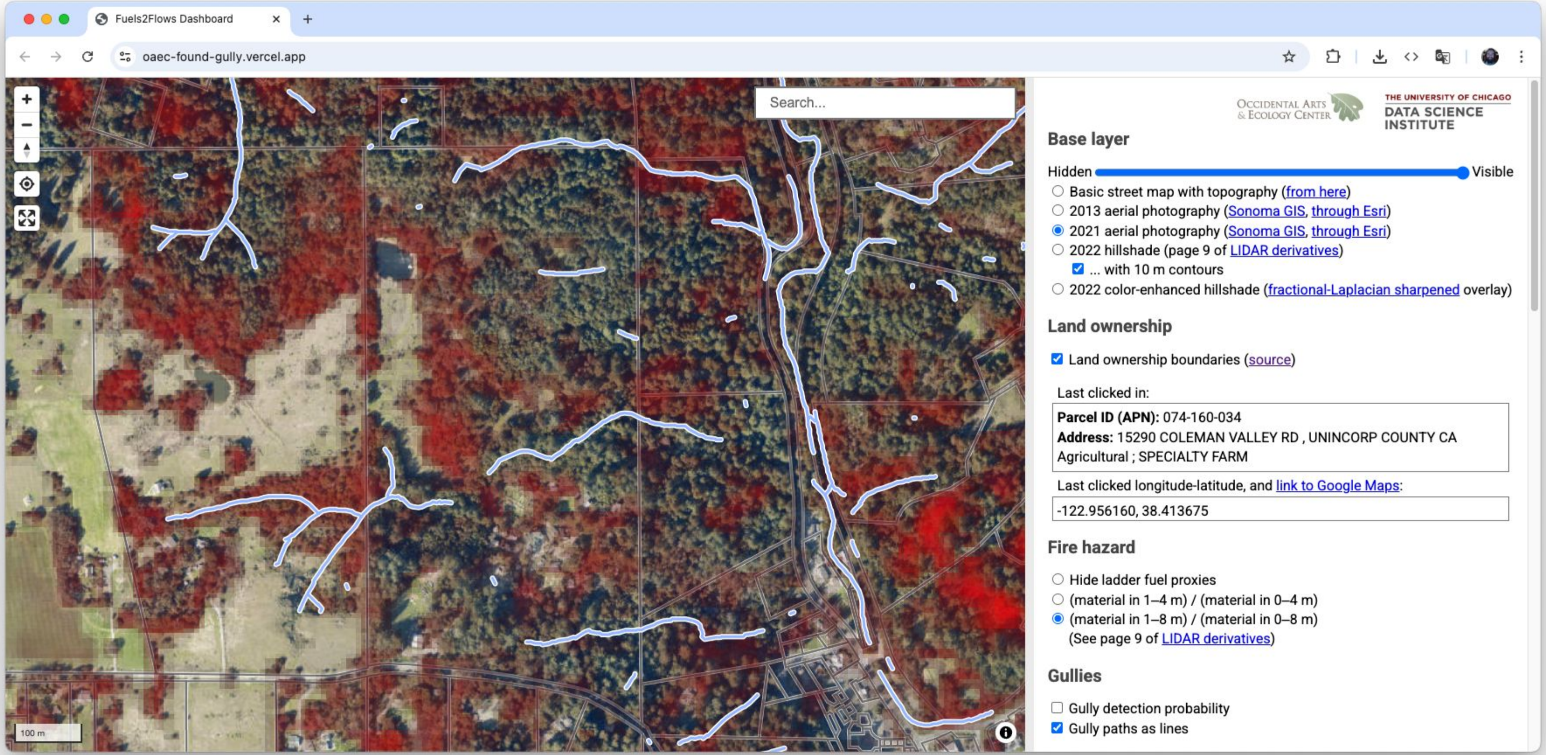
Our final product is a set of map layers

- Gully network
- 2013 and 2021 aerial photography
- 2013 and 2022 elevation data
    - Their difference, which measures erosion and deposition
- Ladder fuel proxies
- Land ownership boundaries and addresses (to contact landowners)

We made a folder of data files available for GIS experts, but

- not all users are likely to be experts;
- some of these files are huge (up to 150 GB).

Instead, we built a specialized map webapp



switch to live demo

Technical highlight

- For performance, the visible layers are delivered on demand as tiles.
- No server required: the PMTiles file format is addressable one tile at a time.
- The "volume of erosion" and "elevation along line" features also use PMTiles to download elevation data one tile at a time.
    - But elevation data needs float32 precision, and PMTiles must be RGBA images.
    - So we bit-cast the four 8-bit channels as a 32-bit float.
    - The (unshown) images look like this:

Impact

This system is for assessing new sites for Fuels to Flows

- We rely, in part, on Brock's network of agencies and landowners who might apply Fuels to Flows on their sites.

- He has been spreading awareness of our site among his contacts.

- We've had two interesting Zoom calls:
  - Adam Cummings at the Watershed Center
  - Matt O'Connor and Jeremy Koor at O'Connor Environmental, Inc.
- and one known user:
  - Fatima Burhan at Sonoma Resource Conservation District

- The gully map is technically complete, but the networking has just begun!

Questions?